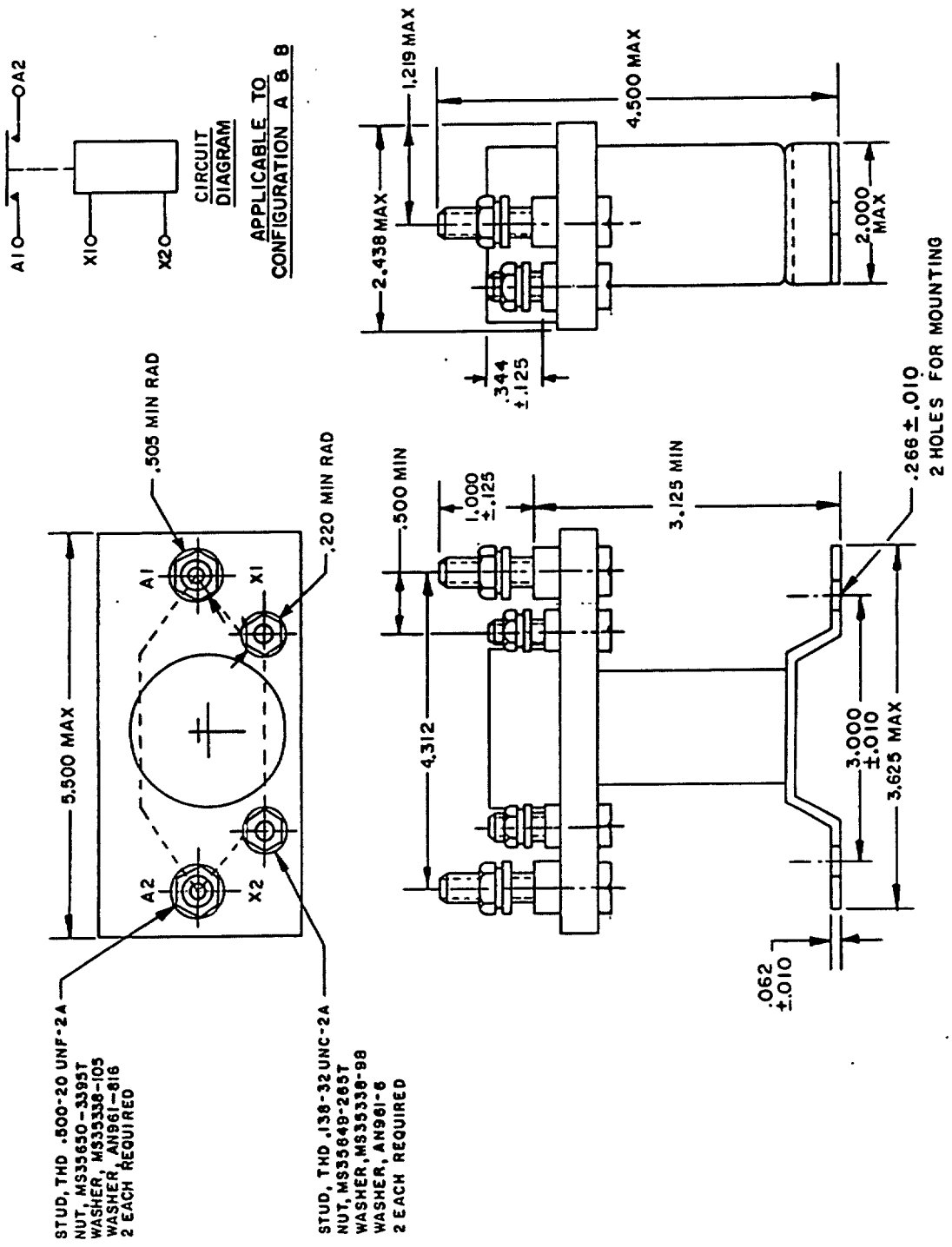
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(K) CONFIGURATON A. Outline drawing (for details see tables I and II).

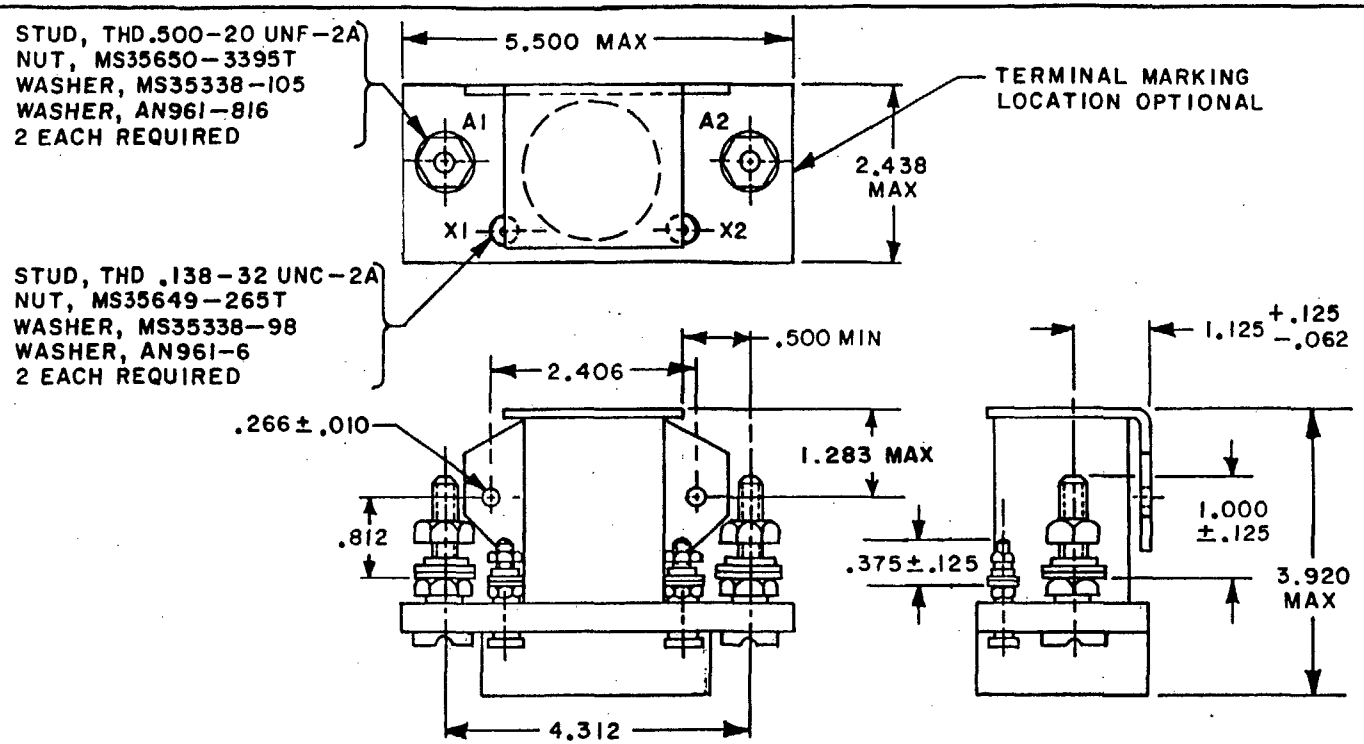
(K) Denotes changes

PREPARING ACTIVITY: Air Force-85 CUSTODIANS: ARMY - NAVY - AS AIR FORCE - DLA - REVIEW: USER: PROJECT NUMBER: 5945-0841-03	MILITARY SPECIFICATION SHEET TITLE RELAYS, ELECTROMAGNETIC, 400 AMPERES, 1 PST, N.O. TYPE II NON-HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER MS24185K 25 July 1990 SUPERSEDING MS24185J 31 Jan 1990 AMSC N/A FSC 5945
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CONFIGURATION B Outline drawing (for details see tables I and II).

Inches	mm	Inches	mm	Inches	mm
.010	0.25	.500	12.70	2.406	61.11
.062	1.57	.505	12.83	2.438	61.93
.090	2.29	.812	20.62	3.000	76.20
.125	3.18	1.000	25.40	3.125	79.38
.138	3.51	1.125	28.58	3.625	92.08
.220	5.59	1.219	30.96	3.920	99.57
.266	6.76	1.273	32.33	4.312	109.52
.375	9.52	1.281	32.54	4.500	114.30
.475	12.06	2.000	50.80	5.500	139.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.062 (1,57 mm). (K)
4. Additional flat washer may be used for terminal seat.
5. Terminal temperature rise under continuous current conditions, 95°C. Intermediate current to be conducted at 71°C.
6. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
8. Shape of relay optional within envelope dimensions.

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TITLE: RELAYS, ELECTROMAGNETIC, 400 AMPERES, 1 PST, N.O. TYPE II NON-HERMETICALLY SEALED

SPECIFICATION SHEET NUMBER: MS24185K
SUPERSEDING: MS24185J 31 Jan 1990
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TABLE I. Dash numbers and characteristics.

Dash number MS24185-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1 1/	II	dc	Stud	Bracket - bottom	2.6
D2 2/	II	dc	Stud	Bracket - side	2.6

1/ For Government logistics support MS24185-D1 shall be used in lieu of AN3380-2.
2/ For Government logistics support MS24185-D2 shall be used in lieu of AN3380-1.

TABLE II. Operating chacteristics.

MS part no. MS24185-	Coil data											Time (milliseconds-maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out Voltage 2/	Operate 3/	Release 4/	Bounce			
		Volts 1/	Freq. Hz	Resn +10% -15% 25°C	Volts	Amperes	Normal 2/	High temp test	Cont. current test					Main		Aux	
														NO	NC	NO	NC
D1	X1, X2	28	dc	66	29	.60	18	21	22.5	7.0	1.5	40	15	5.0			
D2	X1, X2	28	dc	66	29	.60	18	21	22.5	7.0	1.5	40	15	5.0			

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

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TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Main		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	400												
Inductive	10	100												
Inductive														
Motor	50	400												
Lamp														
Transfer load														2/
Mechanical life reduced current	100	100												
Intermediate current	50	40												

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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AMSC N/A FSC 5945

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SPECIFICATION SHEET NUMBER
MS24185K

SUPERSEDING
MS24185 J

31 Jan 1990

AMSC N/A

FSC 5945

Environmental characteristics

Temperature range	-55°C to +71°C
Max altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6 - 9 ms
Max duration contact opening	2 ms

Vibration - Sinusoidal (See table IV)

Vibration - Random N/A

High shock N/A

Acceleration 10 g's

TABLE IV. Vibration levels.

5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
.08 DA	.06 DA	2 g's	2 g's	

Qualification by similarity (See table V)

TABLE V. Qualification by similarity.

	Loads						Dynamics 1/			Environmental		
Part number MS24185-	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	B	C	A	B	C
D1	4						2			4		
D2	4						2			4		

1/ For Group C testing, each dash number shall be tested to meet vibration requirements.

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms
Dielectric strength (sea level)	2-5 seconds

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000 V rms	N/A
Aux contacts	1,250 V rms	N/A	1,000 V rms	N/A
All other points	1,250 V rms	N/A	1,000 V rms	N/A

Dielectric strength (altitude) 1 minute

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	500 V rms	N/A

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	(N.O.) 3200 amperes
Rupture current	(N.O.) 4000 amperes
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

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